

Supplementary Material — Part D

Cross-register coverage

*What Commentators Talk About: A Validated Thematic Taxonomy of Event-Grounded Sports
Commentary*

Transfer of the highlights-derived taxonomy to full-race broadcasts: per-race coverage, and the catalogue of leftover clusters in the residual.

Sections in this part

S6 Per-Race Full-Broadcast Coverage Analysis

S7 Leftover Cluster Catalog (Full-Race Coverage Residual)

S6 — Per-Race Full-Broadcast Coverage Analysis

This supplement provides the full per-race coverage statistics for the ten held-out full-race broadcast transcripts (16,876 sentences; races from 2023–2025). Table 1 reports coverage at the operational threshold $\tau = 0.40$. Table 2 reports coverage sensitivity across $\tau \in [0.30, 0.50]$, together with the number of leftover clusters and the count of clusters that appear in all ten races (universal leftover clusters). Of the 59 leftover clusters at $\tau = 0.40$, 26 appear in all ten races, against a race-label-shuffle null expectation of 37 ± 3 .

Table 1: Per-race full-broadcast coverage at $\tau = 0.40$. Covered = cosine ≥ 0.40 to nearest cluster centroid; Leftover = cosine < 0.40 .

Race	Sentences	Covered	Leftover	Coverage %	Median cosine
2023 Australia	2,466	1,777	689	72.1%	0.538
2023 Qatar	1,336	954	382	71.4%	0.526
2023 Singapore	1,635	1,313	322	80.3%	0.590
2024 Bahrain	1,322	949	373	71.8%	0.519
2024 Brazil	1,905	1,421	484	74.6%	0.542
2024 Italy	1,458	1,101	357	75.5%	0.558
2024 Lasvegas	1,344	1,033	311	76.9%	0.566
2024 Qatar	1,631	1,225	406	75.1%	0.545
2025 Belgium	2,232	1,612	620	72.2%	0.519
2025 Japan	1,547	1,175	372	76.0%	0.554
Total / Pooled	16,876	12,560	4,316	74.4%	—

Table 2: Coverage sensitivity across τ thresholds. Leftover n : uncovered sentences; LC: leftover clusters after re-clustering; Universal: clusters present in all 10 held-out races.

τ	Coverage %	Leftover n	LC total	Universal
0.3	88.8%	1895	8	8
0.35	81.3%	3164	15	11
0.4	74.4%	4316	22	17
0.45	67.2%	5534	21	19
0.5	58.5%	7001	36	26

S7 — Leftover Cluster Catalog (Full-Race Coverage Residual)

This supplement catalogs the leftover clusters identified in the full-race coverage residual (Section 5.4 of the main paper). Each leftover cluster represents a thematic register present in full-race broadcasts but absent or underrepresented in the highlights-derived taxonomy. Universal clusters (present in all held-out races) are marked with a checkmark. The 60 leftover clusters were obtained by re-clustering the

10-race leftover sentences (cosine < 0.40 to any primary cluster centroid) with HDBSCAN using the same hyperparameters as the primary pipeline. Of these, **27** are universal (present in all 10 held-out races). Under a race-label-shuffle null (1,000 permutations of race-session labels across the leftover sentences, leftover partition held fixed) the expected number of universal clusters is 37 ± 3 ; the observed count falls below every permutation drawn ($p < 0.001$), indicating session-specific register structure rather than uniform background noise.

Table 3: All 60 leftover clusters from the full-race coverage residual. Universal: present in all 10 held-out races. Top terms derived by word-frequency from entity-masked sentences.

LC#	Size	Races	Univ.	Top terms
LC0	22	8/10		rule, rules, book, rulebook, allowed, apply, going, record
LC1	23	7/10		sector, middle, sectors, moment, best, got, good, yet
LC2	21	8/10		call, big, bluff, yeah, bluffing, right, let, spade
LC3	36	7/10		commentary, box, louis, replay, time, thought, yeah, going
LC4	52	9/10		medium, soft, hards, hard, mediums, going, softs, set
LC5	65	10/10	✓	weekend, every, haven, going, earlier, week, yeah, far
LC6	32	8/10		grid, boxes, row, take, starts, second, man, today
LC7	44	10/10	✓	top, drive, science, great, drives, way, third, fourth
LC8	97	10/10	✓	radio, engineer, hear, yeah, message, got, going, say
LC9	28	8/10		order, end, establish, running, time, got, right, off
LC10	88	9/10		fans, crowd, always, great, cheer, yeah, hope, big
LC11	20	9/10		move, board, yeah, nice, make, moves, like, great
LC12	32	7/10		nation, time, yeah, going, cup, tea, way, want
LC13	23	8/10		lock, again, front, locking, easier, seen, lockups, locked
LC14	45	8/10		restart, going, order, think, see, procedure, again, reset
LC15	21	9/10		screen, change, red, led, bring, carefully, like, switch
LC16	25	10/10	✓	pass, slam, dunk, passes, going, clean, see, difficult
LC17	45	10/10	✓	field, flat, pole, whole, front, pretty, year, position
LC18	23	6/10		year, years, season, remember, like, form, case, since
LC19	22	9/10		pace, know, got, running, run, good, going, yeah
LC20	49	9/10		start, yeah, good, going, got, rolling, opportunity, best
LC21	22	9/10		result, expect, yeah, fault, end, expecting, expected, sure
LC22	39	10/10	✓	keep, head, lot, got, helmet, hat, sport, everyone
LC23	29	9/10		end, long, way, think, going, try, got, feel
LC24	60	10/10	✓	tough, conditions, difficult, going, hard, easy, know, bit
LC25	50	10/10	✓	season, long, good, year, enough, rookies, start, yeah
LC26	36	8/10		points, today, getting, way, marks, think, going, everything
LC27	43	10/10	✓	sargent, twenty, fight, started, start, birthday, weekend, starting
LC28	46	10/10	✓	seen, year, time, times, like, moment, haven, past
LC29	59	10/10	✓	grip, speed, low, fuel, bit, high, little, corners
LC30	50	9/10		circuit, around, corners, circuits, turn, part, time, area

continued. . .

LC#	Size	Races	Univ.	Top terms
LC31	20	8/10		push, pushing, going, lot, keep, tell, defend, soft
LC32	21	4/10		dry, flat, tarmac, drying, drier, yeah, bit, already
LC33	32	9/10		wide, line, bit, white, yeah, grass, see, lines
LC34	35	10/10	✓	corner, corners, inside, lot, got, make, wall, itself
LC35	33	10/10	✓	temperature, degrees, air, bit, heat, hot, feel, track
LC36	50	10/10	✓	haven, again, away, going, moment, gone, thrown, catching
LC37	25	8/10		gravel, debris, kerbs, like, bit, kerb, trap, surface
LC38	31	10/10	✓	front, left, yeah, side, looking, right, sides, dead
LC39	27	10/10	✓	cockpit, air, clean, little, bit, around, see, got
LC40	30	10/10	✓	teams, year, start, got, bit, something, difficult, haven
LC41	46	10/10	✓	strategy, stop, think, like, different, expecting, done, right
LC42	49	9/10		start, aborted, procedure, formation, right, lap, procedures, investigation
LC43	126	10/10	✓	know, going, risk, think, done, got, looking, need
LC44	24	9/10		inside, pack, air, keep, got, outside, getting, going
LC45	36	10/10	✓	feel, yeah, happy, feeling, right, unwell, like, poised
LC46	76	10/10	✓	going, interesting, think, chance, happen, anything, fascinating, doing
LC47	20	8/10		stint, time, little, think, bit, end, summer, like
LC48	367	10/10	✓	got, think, going, way, time, yeah, done, know
LC49	22	10/10	✓	hear, heard, talk, want, means, talked, love, times
LC50	26	9/10		fun, yeah, lot, stuff, like, ahead, apex, scared
LC51	26	9/10		control, better, bit, yeah, job, think, fell, like
LC52	32	8/10		day, drama, today, dramatic, going, mean, makes, yesterday
LC53	50	9/10		strange, incredible, good, quite, god, got, off, stuff
LC54	180	10/10	✓	yeah, think, know, thing, question, bit, look, little
LC55	60	10/10	✓	close, yeah, surprised, bit, thought, let, know, far
LC56	23	9/10		again, boys, honestly, earlier, sorry, yeah, further, think
LC57	38	10/10	✓	got, going, find, underway, see, good, wonder, let
LC58	59	9/10		let, see, look, yeah, wait, check, again, okay
LC-1	1505	10/10	✓	going, yeah, bit, got, little, think, know, like